USER GUIDE

DANTEC DYNAMICS

Powder Seeding Generator

NOVA INSTRUMENTS

DANTEC DYNAMICS Turn Measurements into Knowledge

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1.0 Introduction

The instruction manual must be read thoroughly before putting the 10F01 Seeding Generator into operation.

The manufacturer accepts no liability for any damage that may occur as a result of improper operation, application, maintenance, nor for operating errors or the use of unsuitable powders.

2.0 Unboxing

Please check that nothing has been damaged during transportation. If the box show signs of cutting or impact, please contact your local Dantec Dynamics representative. Please also check that all parts are included according to the delivery note:

The *10F01 Powder Seeding Generator* includes the following:

- Seeder Vessel
- Seeder Remote Control Unit (Pneumatic)
- Variable voltage supply for seeder vessel.
- Electrical cable for connecting power supply to seeder vessel - 3 meters long.
- Plastic pneumatic tubing (umbilical) connecting Remote Control Panel to seeder vessel - 3meters in length
- Pneumatic tubing (10mm dia 3 meters long) connecting Remote Control unit to regulated air supply.
- Tightening Tool (2 off) for seeder vessel end caps
- User Guide.

If anything is missing, please contact your local Dantec Dynamics representative.

3.0 Safety

3.1 Laser safety

This product is intended for use together with powerful light sources like continuous wave and pulsed laser sources.

The particles delivered by this product are intended to scatter laser radiation for its detection by photosensitive detectors. The scattered radiation is a potential safety hazard as it may damage eyesight if viewed directly. Ensure that experiments are shielded and that all personnel with access to the area exposed to scattered radiation wear appropriate protection.

Read laser supplier’s or laser based instrument suppliers’ safety instructions carefully.

3.2 Working Back-Pressure

The seeder is designed using standard 10 bar pressure vessel components and is certified for operation up to this maximum working back pressure.

3.3 Powder Inhalation

The particles produced by the seeder are very small and since these are easily inhaled, the seed material and operational procedures should also be chosen with regard to toxicity, possible allergic reaction and irritation. The user should refer to relevant local safety recommendations for hazard information and handling/ventilation requirements.

In general, the seeder should be used in well ventilated areas. Forced ventilation is preferred to eliminate seed build up in confined areas. If this cannot be achieved, then personnel should wear appropriate respiratory filters for the powder size range being produced.

Operated at its maximum seed production rate, the device will produce seeding at rates of approximately 800 mm3s-1. For many experiments, seeding is not required continuously and in these cases the total volume of seed material ejected is around 3 liters per hour. This should be borne in mind when considering environmental requirements for filtering exhaust or ventilation flows from experimental areas.

3.4 Static Electricity

Precautions should be taken to avoid static electricity and the risk of explosion. In extreme cases anti-static tubing may be desirable from the output of the seeder to the experiment. The seeder itself is constructed from stainless steel and there are numerous points where earth bonding is possible. Suggested points for any earthing cables are the mounting screws of the pipe clamps that secure the seeder tube to the anodised base-plate.

4.0 Installation and Operating Instructions

4.1 Installation

It is recommended that the seeder is located as close as possible to the rig so that the length of the pipe used to feed the powder into the rig is as short as possible.

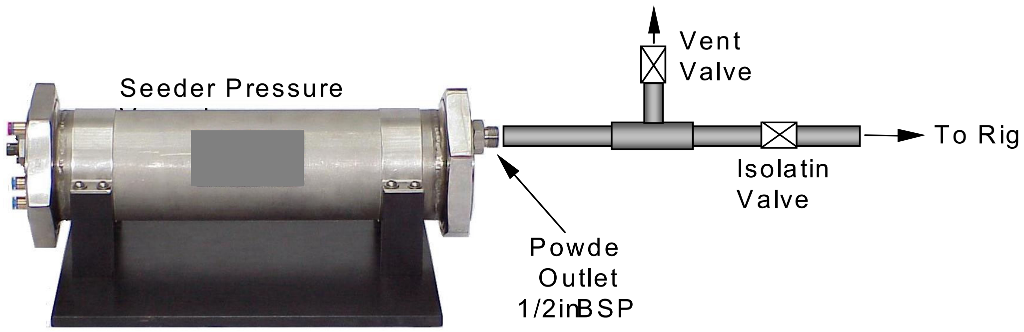

Fig. 1 Recommended installation arrangement. An isolating valve will enable powder refills without shutting down experiment. The vent valve can be used to bleed the pressure inside the seeder to ambient so that the seeder could be safely opened.

The outlet of the Seeder is a high pressure 1/2inch BSP fitting which allows standard fittings to be used. It is also essential that a hand operated valve (not supplied) is fitted between the rig and the seeder so that the seeder can be isolated from the rig and refilled with powder if necessary (see figure 1).

Note: The diameter of the pipe fitted to the outlet should not be smaller than 10mm (approximately) as this will restrict the flow going through the seeder. Without enough airflow flow there is no means by which the powder can be carried out of the seeder tank. The length of the pipe between the seeder and the rig should also be kept short (one meter or below) as this is also likely to reduce the amount of flow through the seeder. Long pipes or pipes with diameter smaller than the recommended sizes will reduce the performance of the seeder.

For high pressure rigs it is also recommended to include a bleed valve so that when the isolating valve is closed the pressure inside the seeder can be vented to ambient before it can be opened safely for a refill. In order to cut down the time it takes for a refill the seeder is supplied with two drums so that the empty one is quickly exchanged with the full drum.

The connections between seeder, the control panel and the variable voltage power supply are shown in figure 2 below. In any case the plastic tubing and the pneumatic connectors are colour coded for convenience.

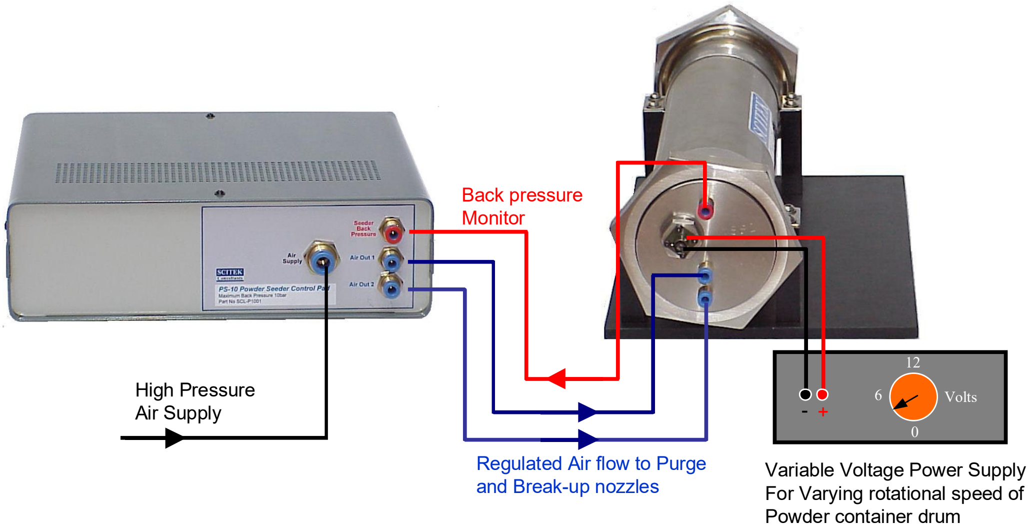

Fig 2. Connection diagram between seeder control pad and variable voltage power supply. Note that the end flange with all the connectors should not be removed unless a major overhaul to the seeder is required. Please consult the supplier before attempting to remove to avoid damage to the seeder.

Note: The flange of the seeder with all the pneumatic and electrical connections should not be dismantled from the seeder unless a major overhaul is necessary. A rubber seal inside this compartment prevents powder from reaching the electric motor and if this is not reassembled properly the motor can be contaminated with powder and become damaged and is likely to fuse the power supply. Please consult the supplier for instructions on how to do this operation to avoid damage to the seeder.

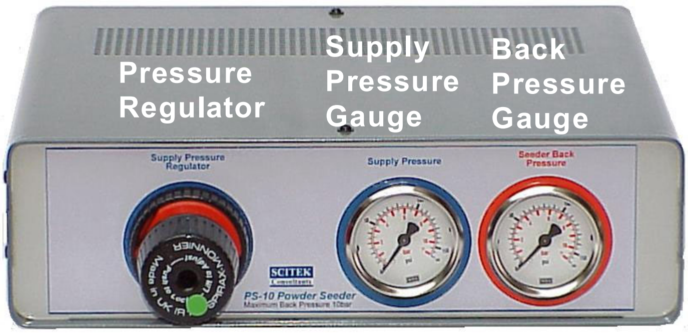

Fig. 3 Front panel of air supply unit. The pressure regulator on this units allows adjustment of the purge and break-up nozzles air supply. The supply pressure is indicated on the blue gauge whereas the red gauge monitors the seeder back-pressure. The supply pressure should always be higher than the back-pressure.

The control pad incorporates a pressure regulator (figure 3) which allows adjustment of the air flow to the seeder for purge and powder break up. The pressure of the air supplied to the seeder can be measured on the pressure gauge adjacent to the pressure regulator knob. Of course this pressure should always be greater than the pressure of the rig (seeder back-pressure) which is indicated on the second pressure gauge on the control pad colour coded red. (Figure 3).

4.2 Filling

1) Ensure that the seeder is isolated from the rig by closing the isolating valve.

2) Ensure that the air supply to the seeder is turned off by unwinding the pressure regulator knob on the control panel or by using the isolating valve on the main feed pipe.

3) Ensure that the drum is not rotating by switching off the variable voltage power supply.

4) Ensure that the pressure inside the seeder is ambient by opening the bleed valve.

5) Remove the outlet flange by unscrewing the 100 mm retaining nut using the “BoaTM” tool supplied.

6) Remove empty drum and wipe any powder off the O-ring seal.

7) Insert spare drum containing fresh powder and ensure that keyed end of the shaft is inserted first and that it engages into the motor drive.

8) Replace the front outlet flange ensuring that the small shaft of the drum is inserted into the bearing on the flange.

9) Replace and hand-tighten the100mm nut, ensuring correct seating of large O ring. Finish tightening with “BoaTM” tool.

10) Increase seeder pressure to the rig pressure by gradually operating the pressure regulator on control panel.

11) Test for leaks and reseal flange if required.

4.3 Operation

1) Fill seeder with powder as above.

2) Turn on main air supply to device, regulated to <10 bar. Ensure that this is higher than the back pressure of the rig. Set air supply to 1.5 or 2.0 times the rig back pressure. (For example if the rig is pressurised to 3 bar then the supply pressure to the seeder should be adjusted to be between 4.5 to 6 bar) The seeder vessel and all pneumatics supplied with it are rated to a maximum of 10 bar operating pressure so ensure that this pressure is never exceeded. For cases where the rig pressure is seven bar the maximum pressure for the air supply to the seeder should be restricted to the 10 bar limit.

3) The seeder is now ready to start dispensing powder and this can be achieved by switching on the power supply to the drum motor, which will start spinning. Always start with the lowest rotational speed (e.g. 2V) and gradually increase it until the required data rate is achieved. The higher the rotational speed the quicker the seeder will use up the powder and more regular refills will be required.

4) In order to minimise powder usage switch unit off when not collecting data. For Aluminium oxide powders of 0.3 microns grain size it is expected that a full drum at 40 rpm rotation (~8V) will last for around 15 to 20 minutes of continuous operation.

4.4 Remote Control Operation

When using powder seeding devices it is essential that the powder dispensing unit is located as close to the rig as possible so that the length of pipe that connects the device to the rig is as short as possible. This ensures that the amount of powder depositing on the walls of the tube (and thus not used as seeding) is minimised.

Powder seeding also has the tendency to coat the rig in which it is used. It is thus desirable to minimise the quantity of powder used and this necessitates a mechanism that allows for powder to be dispensed during data acquisition only. To achieve this the powder seeder incorporates a variable voltage power supply which can be used via an On/Off switch to activate the rotating drum which is the mechanism for dispensing powder. By varying the voltage on this unit it is possible to adjust the rate with which powder is dispensed.

The seeder is also supplied with a remote control unit which allows for the adjustment and monitoring of the supply air pressure to the purge feed and to the break-up nozzles. This unit also incorporates a gauge which monitors the seeder back pressure so that the user can adjust the supply pressure to be higher than the back pressure.

5.0 Reference Guide

5.1 Operating principle

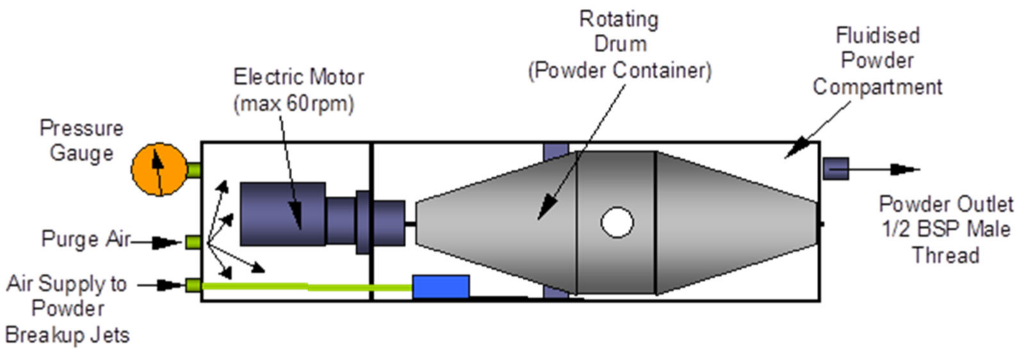

Fig 4. The Powder Seeding Generator and its components.

This device relies on a drum which is filled with powder and contained in a pressure vessel. The drum is rotated about a horizontal axis and at each revolution a small amount of powder is dispensed through a small hole. Of course the rotation of the drum ensures that the powder is agitated which helps to break up big agglomerates. The rate of rotation of the drum is adjustable by means of a variable speed electric motor (figure 4). Thus the rate at which the powder is dispensed is fully adjustable to suit each application. If the drum is not rotated then the seeder seizes to dispense powder and this allows the user to have control as to when powder is dispensed.

Once the powder is dispensed into the pressure chamber a series of sonic air-jets (six in total) are used to break it up into the smallest particles. Baffles that are attached to the outer perimeter of the drum (figure 5) also help agitate heavy agglomerates that remain inside the chamber as they are too heavy to be carried out by the air flow. These baffles also ensure that all the residual powder that gathers inside the chamber in regions where the air flow velocity is low is agitated and carried into the regions of high air velocity where it is broken up by the air jets and dispensed through the outlet.

Fig 5 Powder container drum. Note the grommet which allows the container to be filled with powder and the baffles which agitate agglomerates. The small hole is where the powder is dispensed into the pressure vessel.

In order to prevent the powder from leaking back to the upstream half of the chamber, and contaminating the region where the electric motor is housed, purge air is supplied so that it provides a continuous flow of air from the motor compartment into the powder chamber. Thus powder cannot flow upstream and is forced to remain within the outlet half of the pressure chamber (figure 4).

5.2 Performance

With powders seeders the biggest problem is the breakup of the powder to the smallest grain size. For powders with grain sizes below 1 μm particles tend to agglomerate and unless broken up by the seeder their ability to follow the flow is reduced. The powder seeder employs six sonic jets which create a high shear flow field that breaks up the powder once dispensed by the rotating drum.

The seeder was tested against an oil mist generator in a supersonic wind tunnel and measurements were made of the response of the particles along a normal shock wave using a Laser Anemometer operated in back-scatter. Aluminum Oxide powder with 0.3um size particles was used. As can be seen in figure 6 there is excellent agreement in the results obtained with the two types of seeding devices. The design of this seeder is such that clogging experienced by other powder seeders is avoided. The proprietary powder dispensing method also ensures uniform seeding density during operation.

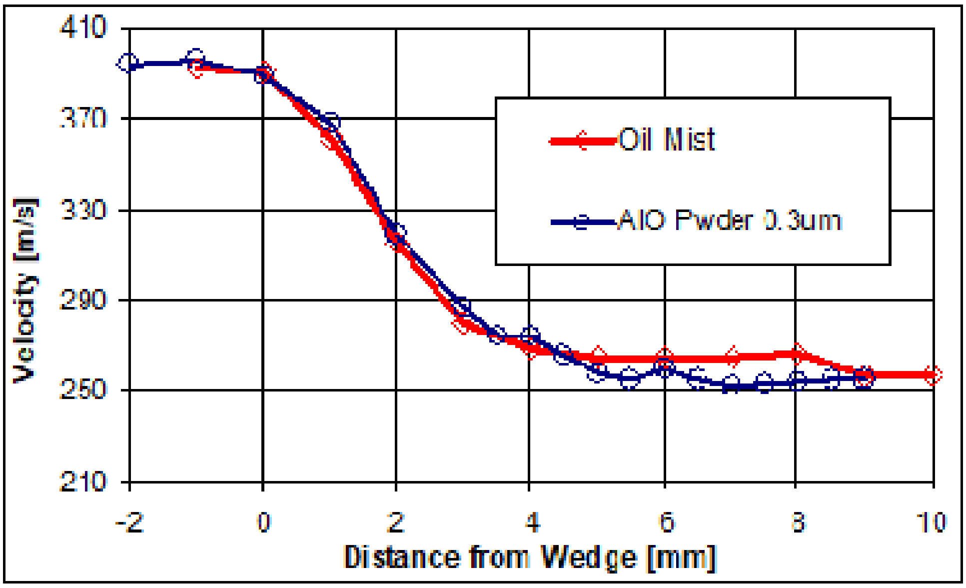

Fig. 6 Comparison of the response of seeding particles in a normal shock. The Powder seeder shows excellent agreement with results from an Oil Mist seeder. Aluminium Oxide powder was used of 0.3 microns in size.

5.3 Recommended Powders

Aluminum oxide powders 0.3 microns in size or higher can be used with this unit. This unit was specifically designed for small size powders which do not readily flow. A full drum at 40 rpm rotation (~8V) it will take around 15 to 20 minutes of continuous operation. Powders of larger grain size tend to flow more readily and for these it is recommended to keep the rotational speed of the drum as low as possible otherwise the powder will be dispersed in a very short time.

For best results it is recommended that the powders are dried in an oven before use and that air supply to the seeder is free from moisture. Moist powder does not easily break up to fine dust and the risk will be the generation of particles that do not follow the flow. The powder containers should also be sealed when in storage to avoid the risk of becoming moist.

6.0 Specifications

- Up to 10 bar working back-pressure.
- Controllable seeding density.
- Dilution air available.
- Stainless steel pressure vessel construction.
- Remote Panel Controls - Electrically operated rate of powder dispersal and thus seed density. Pressure regulator to adjust air supply to seeder. Two pressure gauges to indicate air supply pressure and seeder back pressure. Compatible with solenoid valves for PLC or computer control.
- Push-fit pipe connectors and color coded air tubes for easy installation and adjustment of remote cable length.
- Suitable for a variety of powders and particle sizes. Aluminum oxide or Titanium Dioxide powders are recommended. This seeder was specifically developed for LDA and PIV applications where small particle grain powders need to be used. For LDA applications it is recommended to use Aluminum Oxide powder of 0.3um size, available from us, or from metal polishing suppliers.
- Maximum duration for continuous powder delivery is 30 minutes. A second powder container drum is provided for quick change-over. Remote control panel enables the powder feed to be switched off when not needed thus extending the period between re-fills and minimizing the quantity of powder used.
- Quick re-fill operation without the need to stop experiment.
- Controllability of seeder can be customized to suit other user requirements.

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